

CBT and CBT Apps: Foundations, Fidelity, and Recommendations for an AI-Powered “Build/Run” CBT Protocol App

Foundations and core model of CBT

Cognitive behavioural therapy (CBT) is built around a practical, testable model: how a person interprets events (thoughts/images/meaning-making) is tightly linked to feelings/physiology and to what they do (behaviours), and those behaviours can in turn maintain or worsen distress (for example, avoidance or restriction of activity). ¹ The “change” logic is similarly pragmatic: help the person notice patterns, develop alternative viewpoints/explanations, and then *test* those alternatives in real life (often via behavioural experiments), so that learning generalises beyond the session. ¹

CBT is not just a set of techniques; it is a structured way of working. Core competences described in widely used competency frameworks include working collaboratively, sharing responsibility for session structure/content, setting an agreed agenda, planning and reviewing practice assignments (“homework”), and using summaries and two-way feedback to organise the work. ² Measurement and self-monitoring are also treated as central: select relevant questionnaires/rating scales, establish baselines, use repeated measures to guide treatment decisions, and share progress feedback with the client. ³

A distinctive “in-session” stance in CBT is *guided discovery* (often implemented via Socratic questioning): rather than persuading, the therapist helps the person generate hypotheses, explore evidence, and create opportunities for re-evaluation and new learning. ⁴ This matters for app design because it implies an interaction style (curious, open, non-hectoring) and an epistemology (“let’s test this together”) rather than a didactic coaching voice. ⁵

Active ingredients and process variables linked to successful CBT

Practice between sessions (homework) is one of the most consistently supported process variables in CBT research. A meta-analysis of 23 studies (2,183 participants) found a statistically significant association between homework compliance and treatment outcome (overall $r \approx .26$, small-to-medium). ⁶ Competency frameworks and fidelity scales don’t treat homework as optional: it is framed as the mechanism that integrates session learning with the person’s real environment, with emphasis on clear goals, a rationale, anticipating obstacles, and reviewing what was learned at the next contact. ⁷

Relatedly, CBT is often described as “skills acquisition + generalisation.” That places special weight on (a) making the rationale convincing and personal, (b) making tasks doable, and (c) creating a tight “review → learn → refine next task” loop. ⁸ For an app, this loop is not a nice-to-have; it is one of the most direct translations of “what CBT is” into software mechanics. ⁷

Digital delivery research adds a clear lesson: **support and structure tend to improve outcomes and adherence**, especially for people with higher baseline severity. In an individual-patient-data network meta-analysis of iCBT for depression (36 RCTs; 8,107 analysed participants), guided iCBT was modestly more effective than unguided iCBT at post-treatment (PHQ-9 mean difference about -0.8), and average adherence was higher (about 76% guided vs 54% unguided), with dropout around 25% vs 29% respectively. ⁹ The same analysis found that baseline severity meaningfully modified the added value of guidance, with larger advantages of guidance at higher PHQ-9 scores. ⁹

When iCBT is *guided*, outcomes can be comparable to face-to-face CBT in direct comparisons. A meta-analysis of trials directly comparing guided internet CBT and face-to-face CBT (13 studies; N=1,053) reported an overall post-treatment effect close to zero (Hedges $g \approx -0.01$), consistent with “equivalent overall effects” *in those studied conditions and settings*. ¹⁰ Importantly, many iCBT programmes are explicitly built around modular content that mirrors face-to-face session content and includes psychoeducation, behavioural activation, cognitive restructuring, relapse prevention, and homework. ¹⁰

Engagement is the major bottleneck for consumer mental health apps, particularly unguided ones. A study linking expert app quality ratings with analytics across 56 mental health apps found that user experience scores related to popularity (downloads/MAUs) but **did not strongly predict retention**, and highlighted very low retention as a common pattern in the sector. ¹¹ This creates a design imperative: an app must either (a) achieve sustained engagement, or (b) reliably deliver benefits quickly enough that many users still benefit despite early drop-off. ¹¹

Therapist workflows, fidelity structures, and what “enforcement” looks like in practice

High-fidelity CBT in services is “enforced” through a combination of (1) manualised models, (2) session structure norms, (3) competence frameworks, (4) supervision/training systems, and (5) routine outcome monitoring. ¹² For example, CBT competence lists explicitly include the ability to structure sessions (agenda, homework planning/review, summaries/feedback) and the ability to use measures and self-monitoring to guide therapy and monitor outcomes. ¹³

Within NHS Talking Therapies for anxiety and depression ¹⁴ (stepped-care psychological therapy services in England), delivery is explicitly tied to evidence-based interventions recommended by National Institute for Health and Care Excellence ¹⁵, with low-intensity interventions including guided self-help and *digitally enabled therapy*, and high-intensity interventions including CBT (among others) depending on condition and severity. ¹⁶ The same manual emphasises workforce competence (accredited training, professional registration, and weekly supervision), showing how services operationalise “quality control” beyond simply providing access to a programme. ¹⁷

At the micro-level, fidelity frameworks make the “essentials” visible. In the Revised Cognitive Therapy Scale (CTS-R) style of competence rating, items include collaboration/teamwork, pacing and time management, guided discovery, conceptual integration (a formulation linking triggers/maintaining factors to current work), application of change methods aligned to the formulation, and homework setting with clear goals and rationale. ¹⁸ Even if competence–outcome correlations are not always large, meta-analytic work across psychotherapies indicates that adherence and competence can show small, context-dependent

associations with outcome—enough to justify why systems invest in supervision and fidelity measurement. ¹⁹

For software design, the key translation is: **CBT quality is not just “content correctness”; it is workflow integrity.** In other words, structure (agenda → targeted work → practice plan → review), collaboration cues (shared choices, user agency), and measurement loops (self-monitoring and progress feedback) are part of the treatment. ²⁰

The current CBT app ecosystem and what the evidence indicates

What counts as “CBT app support” varies widely

In the NHS context, “digitally enabled therapies” are typically defined as technologies delivering a substantial portion of therapy content *with support from a trained practitioner/therapist*, who encourages completion, recommends complementary material, and reviews progress and outcomes. ²¹ This definition matters: an app marketed as “CBT” in consumer app stores may deliver CBT-like tools without the support structures that the NHS and much of the iCBT evidence base assumes. ²²

NICE’s view of evidence for CBT-based digital therapies (depression) is selective

In its early value assessment of digitally enabled therapies for adults with depression (updated 2025), NICE lists six CBT-material-based technologies in scope (including Beating the Blues, Deprexis, SilverCloud’s Space from Depression, and Wysa) and concludes that evidence supported recommending some (with further evidence generation) but that evidence was limited or absent for others—explicitly stating limited evidence for Wysa and none for some tools within scope at that time. ²³ The same guidance highlights safety expectations within service protocols, including monitoring and the ability to step care up if symptoms worsen, and flags risks such as suitability for people at higher risk (including suicidal ideation) needing more therapist involvement. ²⁴

Representative examples (approaches, CBT fidelity signals, and evidence)

Product type	Example	How CBT is operationalised (illustrative)	Evidence signal (illustrative)
Digital therapeutic, disorder-specific, high structure	Sleepio ²⁵	A structured self-help programme based on CBT-I; NICE describes access via web/app and optional wearable linking; RCT evidence base includes symptom reduction versus sleep hygiene controls. ²⁶	NICE medical technologies guidance recommends it as an option for insomnia symptoms; large decentralised RCT of SleepioRx showed clinically meaningful ISI improvements sustained to 24 weeks. ²⁷

Product type	Example	How CBT is operationalised (illustrative)	Evidence signal (illustrative)
iCBT platform (often guided/coached)	SilverCloud ²⁸	App/web programmes delivered as modules; in NICE's framing, used with practitioner support; can be implemented as guided vs self-guided iCBT. ²⁹	Used/assessed in multiple studies and service settings; JAMA Psychiatry work describes guided messaging; NICE early value assessment supports some programmes while evidence is generated. ³⁰
Conversational agent delivering CBT-derived content	Woebot ³¹	Short, frequent chatbot conversations, mood check-ins, psychoeducation and CBT skill practice (e.g., cognitive distortions), goal setting, and mood charts; the evaluated system used a decision-tree architecture with embedded NLP rather than open-ended generation. ³²	RCT in a non-clinical student sample found reduced PHQ-9 depression symptoms over 2 weeks compared with an information-only control. ³²
Conversational agent + tool library (CBT/adjacent)	Wysa ³³	Mixed-methods research describes CBT-based tools plus other approaches; notably reports <i>not</i> using AI-generated responses, instead using clinician-created interventive conversations with models routing/understanding. ³⁴	Study of therapeutic alliance suggests users can report alliance-like ratings with a conversational agent; NICE early value assessment (depression) reported limited evidence within its scope at the time. ³⁵
Consumer "AI therapy" / JITAI framing	Youper ³⁶	Observational research frames "AI therapy" as conversational interface + just-in-time adaptive interventions + personalisation; symptom check-ins and repeated emotion-regulation attempts. ³⁷	Large observational study reported early symptom reductions and retention patterns, but explicitly notes limits of non-controlled designs and calls for RCTs. ³⁸
Computerised CBT modules (older generation)	Beating the Blues ³⁹	Multimedia CBT techniques delivered via an interactive programme; historically evaluated in primary care trials. ⁴⁰	Early RCTs found improvements versus usual care, but later pragmatic trials found no additional benefit over usual GP care when engagement was low; NICE later assessed it in digitally enabled therapy evaluations. ⁴¹

Product type	Example	How CBT is operationalised (illustrative)	Evidence signal (illustrative)
Self-help CBT training (web-based)	MoodGYM ⁴²	Web CBT training with workbook/diaries; often used as a self-help programme. ⁴³	Evidence mixed across contexts; REEACT/HTA work reported no added improvement vs usual care in some settings. ⁴⁴
CBT toolkit for anxiety (consumer app)	MindShift CBT ⁴⁵	App store descriptions emphasise CBT tools like thought journaling and a “fear ladder”; feasibility RCT work evaluates acceptability in inpatient contexts. ⁴⁶	Inpatient feasibility RCT suggests acceptability/feasibility constraints; not a definitive efficacy trial for symptom change. ⁴⁷

Two cross-cutting cautions emerge. First, **engagement and adherence** can erase theoretical effectiveness: major pragmatic cCBT trials found low uptake/completion and little added benefit versus usual care in some primary care pathways.⁴⁸ Second, **privacy and security** are not reliably strong across popular mental health apps; an empirical analysis of 27 top-ranked mental health apps found problems such as unnecessary permissions, insecure implementations, and third-party sharing risks that can enable profiling and re-identification.⁴⁹

Evidence-based recommendations for an AI-powered “Build/Run” CBT protocol app

Your proposed concept—an AI-assisted “build mode” that elicits a personalised case understanding and then constructs protocols used in “run mode”—maps well onto recognised CBT competence domains (conceptual integration/formulation + structured session/practice loops).⁵⁰ The core design challenge is to preserve CBT fidelity **without** letting an LLM freely invent therapy content, rationales, or risk decisions beyond evidence-based boundaries.⁵¹

Build mode principles: make the LLM a *formulation and planning assistant*, not a “therapy author”

A CBT-consistent build mode should produce a compact, testable formulation that links triggers → appraisals/thoughts/images → emotions/physiology → behaviours (including avoidance/safety behaviours) → short-term relief and long-term costs, and then selects intervention targets aligned to that map.⁵² In fidelity scales, this is essentially the “conceptual integration” expectation: therapy is guided by a coherent understanding used as the platform for interventions and homework.⁵³

To keep this scientific (and auditable), constrain the LLM to assembling protocols from a **curated library of evidence-based micro-interventions**, each with: indication boundaries, contraindications, steps, example prompts/questions, and “what success looks like.” This mirrors how manualised CBT works in trials and services: a defined method, applied flexibly but not invented ad hoc.⁵⁴

A strong pattern from real-world conversational-agent systems is to avoid open-ended generative responses for clinical content. For example, published research on Wysa reports that it does not use AI-generated responses, instead using clinician-created “interventive conversations” with AI supporting understanding and routing. ⁵⁵ For an LLM-based product, you can still use an LLM—but primarily as an interviewer, summariser, and protocol-parameteriser (e.g., extracting user’s wording for key beliefs, generating candidate behavioural experiments within strict templates, drafting reminders), rather than as the source of therapeutic technique content. ⁵⁶

Run mode principles: implement “micro-sessions” that preserve CBT session integrity

Even if use is ad hoc (“anytime the user needs help”), a run-mode interaction can still follow CBT session structure in miniature:

- **Agenda in one sentence:** “What’s the situation, and what would be most useful to do right now?” (keeps work targeted). ⁵⁷
- **Guided discovery** to elicit the key cognition/meaning and generate alternatives (not debate/persuasion). ⁴
- **Action method aligned to formulation** (cognitive re-evaluation, behavioural experiment, graded task, exposure step, problem-solving step, etc.). ⁷
- **Homework/practice plan** with clear goal, rationale, predicted obstacles, and a prompt to review later. ⁵⁸
- **Summary + feedback** to consolidate learning and calibrate the protocol. ⁵⁷

This structure is not cosmetic: it operationalises the mechanisms supported by CBT competence frameworks and homework-outcome research. ⁵⁹

Measurement-based care: treat metrics as part of treatment, not analytics

CBT competence frameworks explicitly include selecting measures relevant to the client, establishing baselines, using repeated measures to guide therapy, and sharing progress feedback with the client. ³ For digital CBT, measurement also supports stepped-care decisions: NICE’s digitally enabled therapy guidance points to monitoring outcomes and suicidal ideation and stepping treatment up if needed. ²⁴

Concretely, an AI-powered CBT protocol app should: - embed lightweight *in-the-moment* self-monitoring (situation–emotion–behaviour check-ins); ³

- schedule periodic standardised measures where appropriate (e.g., PHQ-9/GAD-7 style symptom tracking is common in trials and iCBT evaluation); ⁶⁰

- use pre-registered decision rules for when protocols are adjusted, paused, or escalated to human support. ⁶¹

Personalisation: use a JITAI lens, but anchor it to CBT tasks

Your “run mode anytime” framing naturally aligns with *just-in-time adaptive interventions* (JITAI): delivering the right kind/amount of support at the right time based on changing context/state. ⁶² Published work on Youper explicitly frames “AI therapy” around just-in-time interventions and adaptation/personalisation, and empirically links repeated “successful regulation attempts” with symptom change (observationally). ³⁷

A CBT-faithful JITAI implementation would adapt **which CBT protocol** is offered and **what step** is suggested next (e.g., next exposure task rung, next behavioural activation activity), while keeping the *protocol's internal logic* manualised and auditable. ⁶³

Human support as a product feature, not an optional extra

Across iCBT evidence, guidance improves adherence and can improve short-term outcomes, particularly at higher severity levels. ⁶⁴ NICE's definition of digitally enabled therapy for depression in NHS Talking Therapies explicitly assumes practitioner support that encourages completion and reviews progress/outcomes. ²¹

For an AI-powered CBT app, the most evidence-aligned approach is therefore **a stepped-support design**: - default: automated, structured CBT protocols;
- add-on: optional coach/therapist messaging, or escalation paths triggered by deterioration, non-engagement, or risk signals. ⁶⁵

This is not only clinically motivated; it is also a predictable route to better engagement in a sector where retention is generally poor. ⁶⁶

Safety boundaries: build escalation and crisis routing into the protocol engine

NICE's digitally enabled therapy guidance explicitly raises suitability issues for people at higher risk (including suicidal ideation) and emphasises monitoring and the possibility of treatment increase/alternative intervention. ⁶⁷ A consumer-facing AI CBT app should implement: - clear "not for emergencies" boundaries,
- active detection of crisis signals, and
- immediate routing to UK crisis pathways (e.g., NHS urgent mental health help guidance). ⁶⁸

Evaluation, governance, and UK deployment requirements for an AI CBT app

An AI-powered CBT protocol builder is likely to be treated by regulators and health systems as more than a wellness app if it makes treatment claims, adapts interventions using health data, or affects clinical decisions. The Medicines and Healthcare products Regulatory Agency ⁶⁹ notes that many software products (including AI) are regulated as medical devices in the UK depending on intended purpose and claims. ⁷⁰

In the UK commissioning/evaluation ecosystem, two frameworks are especially relevant:

- NICE's Evidence Standards Framework (ESF) for digital health technologies describes evidence expectations, including design factors and deployment considerations, and stratifies evidence requirements by risk tier; it explicitly discusses adaptive algorithms whose performance may change over time. ⁷¹
- NHS England's Digital Technology Assessment Criteria (DTAC) sets baseline expectations for technologies entering NHS/social care procurement contexts (clinical safety, data protection, technical security, interoperability, usability/accessibility). ⁷²

NICE's ESF also explicitly points to compliance areas relevant to a mental health AI app: regulatory marking (e.g., UKCA as applicable), data protection law (GDPR as implemented in UK law), information governance, and DTAC-related assurance. ⁷³

For mental health apps specifically, evaluation should not be limited to symptom change. The American Psychiatric Association ⁷⁴ provides an app evaluation model that emphasises hierarchical assessment categories (including privacy/security, evidence, usability, and data integration considerations), reflecting the reality that app selection is not the same as selecting a traditional psychotherapy. ⁷⁵

Finally, privacy and data security are not abstract concerns in this domain. Empirical security/privacy analysis of popular mental health apps found risks including unnecessary permissions, insecure implementations, and third-party sharing that can enable profiling and re-identification—especially sensitive given the stigma and potential harms linked to mental health data exposure. ⁴⁹ These findings support a “privacy by design” posture as a core product requirement, not a compliance afterthought. ⁷⁶

Selected short quotes

“The average adherence was 76% for guided iCBT and 54% for unguided iCBT.” ⁹

“Digitally enabled therapies ... are delivered with support from a trained practitioner or therapist.” ²¹

“The aim of helping clients ... use behavioural experiments to test-out the accuracy of these alternatives.” ¹

“The style of the therapist ... should be open and inquisitive.” ⁵³

¹ ² ³ ¹³ ²⁰ ³⁶ ³⁹ ⁵² ⁵⁷ https://www.ucl.ac.uk/brain-sciences/sites/brain_sciences/files/migrated-files/Basic_CBT_Competences.pdf

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